

KUALA TRENGGANU, Malaysia

WMO: 486180

Lat: 5.3827N

Lon: 103.1080E

Elev: 21

StdP: 14.68

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	71.5	73.0	67.9	102.6	76.3	69.4	108.2	76.8	16.5	79.0	15.0	79.8	3.4	200	0.328

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	13.2	91.8	81.9	91.6	81.8	91.1	81.6	83.5	89.9	82.8	89.4	82.2	89.0	6.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.0	161.4	88.3	80.8	160.3	88.2	80.5	158.6	88.0	47.4	90.6	46.7	90.1	46.1	88.9	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.9	11.2	9.8	70.2	92.8	1.7	1.3	68.9	93.8	68.0	94.5	67.0	95.3	65.8	96.2		
			68.5	85.1	1.8	1.2	67.2	86.0	66.2	86.7	65.2	87.4	63.9	88.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.9	80.0	80.8	81.8	83.3	83.8	83.3	82.7	82.5	82.2	81.6	80.7	80.3
	DBStd	2.15	2.01	1.82	1.82	1.83	1.84	1.77	1.70	1.51	1.37	1.70	1.94	1.93
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11653	931	863	986	998	1049	999	1014	1008	966	978	921	940
	CDD65	6178	466	443	521	548	584	549	549	543	516	513	471	475
	CDH74	64546	4407	4532	5659	6416	6793	6081	5860	5723	5257	5014	4329	4475
	CDH80	23026	999	1320	2046	2687	2907	2502	2296	2227	1998	1719	1258	1067
Wind	WSAvg	4.4	5.8	5.3	4.5	4.1	4.0	3.9	3.8	3.9	3.9	3.9	4.1	5.9
Precipitation	PrecAvg	100.7	6.1	3.1	3.8	3.9	4.1	4.1	4.2	5.3	6.2	10.7	26.6	22.6
	PrecMax	180.8	39.4	14.1	17.0	16.1	13.9	9.0	10.2	11.4	13.3	33.9	62.8	54.2
	PrecMin	67.9	0.5	0.0	0.0	0.0	0.2	0.9	0.9	1.3	0.9	3.6	8.9	2.8
	PrecStd	24.4	8.9	3.2	3.8	3.4	3.2	2.1	2.2	2.6	3.1	6.4	15.1	14.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.1	88.3	91.2	93.1	93.5	93.2	91.8	91.6	91.5	90.7	89.7	88.2
		MCWB	80.0	79.3	80.2	82.7	82.8	82.5	82.0	81.0	81.4	81.0	80.7	80.1
	2%	DB	86.4	87.9	89.8	91.6	92.8	91.7	91.3	90.0	89.9	89.7	88.1	87.6
		MCWB	79.1	79.2	79.8	82.0	82.5	81.8	81.6	81.0	80.5	80.5	80.5	79.9
	5%	DB	85.9	86.4	89.1	91.1	91.5	91.2	89.9	89.7	89.6	88.2	87.6	86.1
		MCWB	78.7	78.3	79.6	81.7	82.0	81.6	80.9	80.9	80.4	80.2	80.3	78.9
	10%	DB	84.4	86.0	87.8	89.7	91.0	89.7	89.3	89.2	88.1	87.7	86.1	84.5
		MCWB	77.7	78.1	79.4	80.7	81.7	80.9	80.7	80.8	80.2	80.0	79.4	78.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.7	80.7	82.4	84.1	83.7	83.9	83.7	84.1	82.8	82.8	81.9	81.0
		MCDB	85.2	86.9	88.3	90.6	91.6	90.4	90.1	89.3	88.9	88.0	86.6	85.6
	2%	WB	80.2	79.9	81.0	82.7	83.1	83.1	82.6	82.6	81.9	81.5	80.9	80.2
		MCDB	84.9	86.0	87.5	90.2	90.9	89.6	89.0	88.6	88.4	87.5	86.2	85.0
	5%	WB	79.5	79.4	80.3	81.8	82.4	82.4	81.8	81.7	81.1	80.7	80.3	79.7
		MCDB	84.0	85.2	86.7	89.2	90.0	88.9	88.1	87.9	87.7	87.1	85.5	84.1
	10%	WB	78.7	78.7	79.7	81.3	81.7	81.6	81.0	80.9	80.3	80.0	79.8	79.0
		MCDB	83.0	84.3	85.8	88.4	88.9	88.0	87.3	87.3	86.7	85.9	84.7	83.2
Mean Daily Temperature Range	5% DB	MDBR	8.8	10.4	11.6	12.6	13.2	13.3	13.2	13.3	13.2	11.8	10.1	8.6
		MCDBR	10.4	11.1	13.1	13.7	14.2	14.1	14.0	13.9	13.9	13.1	12.0	10.4
		MCWBR	4.9	4.9	5.4	6.0	6.4	6.7	7.1	7.1	6.9	6.2	5.6	5.0
	5% WB	MCDBR	9.2	10.2	11.9	13.1	13.6	13.3	13.4	13.4	13.4	12.6	11.1	9.6
		MCWBR	5.1	4.8	5.5	6.2	6.7	7.0	7.3	7.6	7.1	6.3	5.5	5.0
Clear-Sky Solar Irradiance	taub	0.465	0.452	0.458	0.439	0.458	0.480	0.489	0.494	0.495	0.485	0.447	0.475	
	taud	2.306	2.328	2.329	2.383	2.340	2.287	2.256	2.226	2.224	2.260	2.383	2.289	
	Ebn at Noon	272	278	276	277	265	256	254	258	262	266	275	266	
	Edh at Noon	42	42	43	40	40	42	43	46	47	45	39	43	
All-Sky Solar Radiation	RadAvg	1376	1714	1837	1833	1676	1611	1588	1653	1659	1455	1263	1195	
	RadStd	121	135	153	94	88	74	91	45	86	89	93	119	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	+0.36	N/A	N/A	+0.54	+0.94	+0.81	N/A	N/A	+117

Nomenclature: See separate page